Raw Signals, Handcrafted Features, or Learned Embeddings? A Controlled Comparison of Eye-Tracking Representations for User-State Prediction

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Abstract

Eye tracking can provide behavioural evidence about user state, but it remains unclear which representation is most useful when predictions must generalise to unseen people. This paper compares four representations of Tobii eye-tracking windows from the TrustMe study: raw gaze and pupil signals, handcrafted features, GazeMAE embeddings, and MOMENT embeddings. We frame the task as predicting the provisional self-report target *q5* and evaluate a majority baseline, Random Forest, and multilayer perceptron under a leave-one-subject-out protocol with fold-local preprocessing. We report accuracy, balanced accuracy, macro-F1, and AUC, retaining undefined AUC values for one-class held-out folds. [To complete: Insert the common cohort size, the principal quantitative result, and the resulting practical recommendation.]

Keywords

eye tracking, user-state prediction, representation learning, GazeMAE, MOMENT, subject generalisation

1 Introduction

Eye tracking offers a detailed record of gaze behaviour and pupil dynamics that can support inferences about user state in interactive systems [Citation needed]. Yet, choosing how to represent those signals remains a practical and methodological decision. Raw windows retain temporal detail; handcrafted features encode established domain knowledge and can be easier to interpret; and learned embeddings may transfer useful structure from gaze-specific or general time-series pretraining.

This study makes that choice explicit through a controlled comparison on Tobii windows from the TrustMe study. The same labelled windows and subject- generalisation protocol are used to compare raw signals, handcrafted features, GazeMAE embeddings, and MOMENT embeddings. The provisional primary target is *q5*; its wording, scale, and substantive interpretation are documented in Section 3 before the final submission.

Our research question is: *Which eye-tracking representation best predicts the selected TrustMe self-report state for unseen subjects?* We make three contributions: (1) a like-for-like comparison of four representation families; (2) a leakage-safe leave-one-subject-out evaluation with fold-local preprocessing; and (3) evidence-based guidance on the performance, interpretability, and implementation trade-offs of each representation.

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2 Related Work

[To complete: Review eye tracking for user-state, engagement, cognitive-load, and affect prediction; introduce classical gaze and pupil features; and cite the original GazeMAE and MOMENT papers. Position this work as an empirical comparison, not a new representation-learning method.]

Subject-dependent behavioural data can give overly optimistic results under random window-level splits. [To complete: Add literature supporting subject-disjoint evaluation and fold-local normalisation.]

3 Dataset and Preprocessing

3.1 TrustMe Tobii Data

The analysis starts from already computed representations of TrustMe Tobii recordings. Raw participant data, derived data, model weights, and results are not included in this repository. [To complete: State the study purpose, ethics or consent information where applicable, participant count, recording setup, sampling characteristics, session structure, and exclusions.]

3.2 Windows and Target

Representations are aligned through a common window identifier (`window_uid`). Supervised analyses use labelled windows only and use their common intersection across the four representations, ensuring that a representation is not advantaged by a different cohort. [To complete: Specify window duration, overlap, ordering within participant, the exact q5 question, response scale, binarisation rule, and final counts of subjects and windows.]

3.3 Data Quality and Fold-Local Processing

Raw Tobii channels comprise left and right pupil size, horizontal and vertical gaze coordinates, and average distance. Validity fields are used during raw signal processing. For the main protocol, all data-dependent operations, including imputation, feature filtering, scaling where enabled, and target centring, are fitted using training subjects only and then applied to the held-out subject. [To complete: Report the final missing-data policy and quality-control exclusions.]

4 Eye-Tracking Representations

We compare the following families in a fixed order throughout the paper.

4.1 Raw Signals

Each raw window contains five channels: left and right pupil size, gaze point coordinates (x, y), and average distance. The current experimental configuration uses 120 samples per channel with truncate-or-pad length handling. [To complete: Confirm the final preprocessing, input dimensionality, and whether this configuration is retained for the submitted experiment.]

4.2 Handcrafted Features

The handcrafted representation is loaded from the TrustMe feature export. Structural metadata are excluded; features with excessive missingness and highly correlated features are filtered within each training fold. [To complete: List the retained feature families and the final number of features.]

4.3 GazeMAE Embeddings

GazeMAE provides a gaze-specific learned embedding for each window [1]. [To complete: Document the checkpoint, its input signals, pooling or embedding selection, quality-control requirements, and final dimensionality. Cite the source paper.]

4.4 MOMENT Embeddings

MOMENT provides a general-purpose time-series embedding of each window [2]. [To complete: Document the pretrained model, input channels, pooling method, quality-control requirements, final dimensionality, and any window-count difference relative to GazeMAE. Cite the source paper.]

5 Experimental Setup

5.1 Models

We use a most-frequent-class classifier as the majority reference baseline, a class-balanced Random Forest with 300 trees, and a multilayer perceptron. The current MLP configuration has hidden layers of sizes 1028, 512, 256, 128, and 64, GELU activations, Adam optimisation, early stopping, and a fixed seed. [To complete: Confirm hyperparameters after the final run and explain any model-specific preprocessing.]

5.2 Primary Evaluation Protocol

The primary analysis is leave-one-subject-out (LOSO) evaluation with the normalised label rule configured in the project. Each fold holds out one participant, preventing information from that person's windows from entering training or preprocessing. The current q5 rule centres targets using training subjects and handles an unseen subject using the global training mean, with a threshold of 3.0. [To complete: Verify the interpretation of this rule and state the exact final target definition in reader-facing language.]

5.3 Metrics and Reporting

We report accuracy and balanced accuracy with their deltas over the majority baseline, macro-F1, and AUC. One-class held-out folds are retained; metrics that are undefined for such a fold are recorded as `NaN` rather than silently discarded. Primary summaries aggregate at the subject level. [To complete: Specify uncertainty intervals or paired statistical comparisons, if used.]

6 Results

[To complete: Do not draft conclusions here before the final experiment run. Insert a main table with representation, model, accuracy, Δ accuracy, balanced accuracy, Δ balanced accuracy, macro-F1, AUC, and subject-level uncertainty. Order representations as raw, handcrafted, GazeMAE, and MOMENT; show the majority baseline first.]

6.1 Main Representation Comparison

[To complete: State the principal LOSO result, compare it with the majority baseline, and distinguish representation effects from classifier effects.]

6.2 Subject-Level Results

[To complete: Add a per-subject performance figure or distribution and row-normalised confusion matrices for the selected comparisons. Use a fixed colour scale from 0 to 1 and discuss one-class folds.]

7 Discussion

[To complete: Answer the research question using the final results. Discuss whether learned embeddings improve on raw signals or handcrafted features; separate performance from preprocessing cost, interpretability, and deployment constraints; and avoid treating two-dimensional projections as proof of class separability.]

7.1 Limitations

The conclusions are limited to one dataset, one windowing configuration, a provisional self-report target, and the selected pretrained checkpoints. [To complete: Add final limitations concerning sample size, label noise, quality-control decisions, and generalisability.]

8 Conclusion

[To complete: Summarise the final controlled comparison, its strongest result, and a concise recommendation for practitioners choosing an eye-tracking representation.]

Acknowledgements

[To complete: Add funding, institutional, and data-collection acknowledgments only if required and approved.]

References

[1] Louise Gillian C. Bautista and Jr. Naval Prospero C. 2020. GazeMAE: general representations of eye movements using a micro-macro autoencoder. *arXiv preprint arXiv:2009.02437*. https://arxiv.org/abs/2009.02437.

[2] Mononito Goswami, Konrad Szafer, Arjun Choudhry, Yifu Cai, Shuo Li, and Artur Dubrawski. 2024. MOMENT: a family of open time-series foundation models. *arXiv preprint arXiv:2402.03885*. https://arxiv.org/abs/2402.03885.

A Reproducibility Details

[To complete: Add the exact YAML configuration, representation dimensions, common-intersection counts, software versions, per-fold metrics, and secondary protocol or projection results that do not fit in the main paper.]